Human Resources Dataset Analysis

1. Project Planning & Management

1.1 Project Proposal

Overview. This project analyzes a Human Resources dataset to uncover what drives employee satisfaction, performance, and attrition, and to forecast workforce trends. The work follows a full analytics lifecycle: data modeling and cleaning, analytical question design, forecasting, and a Tableau dashboard with a final report and presentation.

Objectives

- Build a clean data model and a cleaned, analysis-ready HR dataset.
- Define and answer the analysis questions that matter to HR decision makers (for example, the relation between employee age and satisfaction level).
- Answer a set of forecasting questions using trends found in the data.
- Deliver an interactive Tableau dashboard that visualizes the answers to all questions.
- Deliver a final report and presentation covering analysis, model development, and deployment.

Scope

- In scope: data modeling, cleaning and preprocessing, exploratory analysis, forecasting with scikit-learn, Tableau dashboard, report, presentation, GitHub documentation.
- Out of scope: collecting new data, building a production web application, real-time data pipelines.

1.2 Project Plan

Timeline (4 weeks)

Phase	Week 1	Week 2	Week 3	Week 4
Data model, cleaning and preprocessing	■			
Analysis questions and EDA		■		

Phase	Week 1	Week 2	Week 3	Week 4
Tableau dashboard				
Final report and presentation				

Milestones and deliverables

Milestone	Week	Deliverables	Tools
M1: Data Preprocessing & Modeling	1	Cleaned dataset, data preprocessing notebook, SQL data model (schema/ERD)	Python (pandas), SQL for modeling
M2: EDA	2	EDA notebook with findings and visuals.	Python (pandas, Matplotlib)
M3: Business Questions	3	Set of business questions answered with SQL queries and results.	SQL
M4: Dashboard & Final Delivery	4	Tableau dashboard, final report, final presentation	Tableau, MS PowerPoint

Resource allocation

- Software: Python with Jupyter Notebook, SQL database, Tableau, MS PowerPoint, GitHub.
- Data: the provided HR dataset.
- People: see 1.3.

1.3 Task Assignment & Roles

Role	Responsibility	Assigned to
Project lead	Planning, tracking milestones, final submission	Hazem
Data Cleaning & Modeling	Provide clean dataset	Hazem
Data Modeling	Design the SQL data model (tables, keys, relationships, ERD)	Omnia, Hazem
Data analysis	EDA	Eyad
Business Question Answering	Define business questions and answer them with SQL	Ahmed
Data Visualization	Tableau dashboard	Rehab
Documentation	GitHub Rep. and project documents	Team Contribution
Final Report & Presentation	Final report, final presentation	Team Contribution

1.4 Risk Assessment & Mitigation Plan

#	Risk	Likelihood	Impact	Mitigation
R1	Missing, inconsistent, or duplicated values in the dataset	High	Medium	Profile data early; document every cleaning rule in the notebook; keep the raw file untouched

#	Risk	Likelihood	Impact	Mitigation
R2	Sensitive employee attributes (age, gender, salary)	Medium	High	Use only the provided dataset; avoid identifying individuals; report aggregated results
R3	Time pressure at the Week 4 dashboard and report stage	Medium	High	Draft report sections during Weeks 1 to 3; start dashboard sketches in Week 2
R4	Tool problems (Tableau licensing, environment issues)	Low	Medium	Keep a requirements file for Python; keep a backup export of dashboard data as CSV
R5	Loss of work or version confusion	Low	High	Commit to GitHub regularly with meaningful messages

1.5 KPIs (Key Performance Indicators)

KPI	Target
Data quality	100% of columns profiled; missing values and duplicates resolved or documented
Milestone delivery	All 4 weekly deliverables completed on schedule
Analysis coverage	Every analysis question has a documented answer and a visual
Forecast quality	Each forecasting question answered with a plot and an evaluation metric (for example MAE or RMSE)
Dashboard completeness	Every answered question is represented in the Tableau dashboard
Documentation	All required documents uploaded to GitHub before the deadlines

2. Literature Review

2.1 Background and Related Work

HR analytics applies data analysis to workforce questions. Published work and industry practice on this topic cluster around a few recurring themes that this project builds on.

- **Employee attrition and retention.** Most studies treat attrition as a prediction or driver-analysis problem, relating turnover to factors such as satisfaction, workload, tenure, compensation, and promotion history.
- **Satisfaction and engagement.** Research commonly examines how demographics (such as age and tenure) and work conditions relate to satisfaction scores, usually through descriptive statistics, correlation, and group comparisons.
- **Communicating results.** Dashboards are the standard way to give HR decision makers self-service access to findings.

Source: https://www.emerald.com/bij/article/31/9/3102/1222050/Role-of-HR-analytics-and-attrition-on

2.2 How This Project Fits

Theme	Application in this project
Driver analysis	Analysis questions relating employee attributes to satisfaction and other outcomes
Data preparation	SQL and pandas cleaning pipeline with a documented data model
Communication	Tableau dashboard and final presentation

2.3 Feedback & Evaluation

To be completed by the instructor.

Criterion	Comments
Clarity of objectives	
Quality of data work	
Analytical depth	
Presentation of results	

2.4 Suggested Improvements

Areas where the project can be enhanced if time allows:

- Add statistical tests to support key findings.
- Add dashboard filters (department, age group, tenure) for self-service exploration.
- Document dataset limitations and bias risks in more detail.

2.5 Final Grading Criteria

Component	Marks
Documentation	
Implementation	
Testing	
Presentation	

3. Requirements Gathering

3.1 Stakeholder Analysis

Stakeholder	Role	Needs
HR managers	Primary decision makers	Understand what drives satisfaction and attrition; see trends and forecasts
Executives / leadership	Strategic oversight	High-level KPIs and clear summaries
Department heads	Team management	Department-level comparisons and insights
Data analyst (project author)	Builds and maintains the analysis	Clean data, reproducible code, clear requirements
Instructor / evaluators	Assessment	Complete documentation, working deliverables, GitHub repository

3.2 User Stories & Use Cases

User stories

- As an HR manager, I want to see how employee age relates to satisfaction level so that I can target engagement programs.
- As an HR manager, I want to see which departments or groups have the highest attrition so that I can prioritize retention efforts.
- As an executive, I want a single dashboard with the key findings so that I can review results quickly.
- As an HR planner, I want forecasts of workforce trends so that I can plan hiring and budgets.
- As an evaluator, I want a documented, reproducible repository so that I can verify the work.

Use cases

ID	Use case	Actor	Description
UC1	View dashboard	HR manager, executive	Opens the Tableau dashboard and reviews the answers to all analysis questions
UC2	Filter and explore	HR manager	Filters views by attributes such as department or age group
UC3	Reproduce analysis	Evaluator	Clones the GitHub repo and runs the notebook following the README

3.3 Functional Requirements

ID	Requirement
FR1	The system shall load the HR dataset and build a documented data model.
FR2	The system shall clean and preprocess the data (missing values, duplicates, data types, inconsistent categories) and output a cleaned dataset.
FR3	The project shall provide a data preprocessing notebook documenting every cleaning step.
FR4	The project shall define a set of analysis questions of interest to HR decision makers and answer them using SQL.
FR5	The project shall provide a Tableau dashboard that visualizes the answers to all answered questions.
FR6	The project shall provide a final report and a final presentation covering data analysis, model development, and deployment.
FR7	All documents and code shall be uploaded to GitHub with a README.

3.4 Non-functional Requirements

Category	Requirement
Performance	Notebook and dashboard load and run in reasonable time on a standard laptop
Usability	Dashboard is clear, consistently labeled, and understandable by non-technical HR staff
Reliability	Analysis is reproducible: same input data produces the same outputs

Category	Requirement
Security and privacy	No personal identifiers exposed; results reported in aggregate
Maintainability	Code is modular, commented, and follows consistent naming conventions
Documentation	Project documents and code are uploaded to GitHub with a README